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RESEARCH ARTICLE

Deepfake Audio Detection: A Comparative Study of Advanced Deep Learning Models

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ABSTRACT The rapid advancements in artificial intelligence (AI) have significantly enhanced audio synthesis capabilities, enabling deepfake technology to replicate human speech with near-perfect accuracy. This poses severe security threats to various domains, including banking, customer service, and law enforcement, as malicious actors can exploit speech synthesis techniques such as voice conversion, replay attacks, and text-to-speech (TTS) to manipulate or impersonate individuals. In this study, both spectral and temporal features were extracted from audio recordings to explore state-of-the-art deepfake audio detection frameworks, highlighting the effectiveness of advanced deep learning methods. Specifically, the performance of the bidirectional long short-term memory (BLSTM) network, a custom convolutional neural network (CNN), a residual CNN integrated with an attention mechanism and bidirectional gated recurrent unit (ResCNN-Attention-BGRU), WIREnet (a variant of BLSTM), the residual network for fully connected (ResNet FC), and a squeeze-and-excitation-enhanced one-dimensional CNN (SE-Enhanced 1D-CNN) were compared. The proposed models achieved notable testing accuracies, with SE-Enhanced 1D-CNN reaching the highest at 97.64%, followed by ResNet FC (97.46%) and WIREnet (97.20%), demonstrating strong generalization across the evaluated architectures. Furthermore, we experimented with different numbers of mel-frequency cepstral coefficients (MFCCs), specifically 13, 26, and 39, in combination with other spectral and temporal features. The experimental results demonstrated that MFCC-39, together with spectral and temporal features, had a robust feature representation and achieved the best performance for deepfake detection.

INDEX TERMS Deepfake audio, artificial intelligence, convolutional neural network, mel-frequency cepstral coefficients, SE-enhanced 1D-CNN, bidirectional long short-term memory network.

I. INTRODUCTION

A Deepfake audio, generated or manipulated using AI, has seen a significant rise in applications associated with fraudulent activities, including identity theft and financial scams. The rapid advancements in AI-driven audio synthesis have exacerbated security vulnerabilities, particularly in voice-based authentication systems, rendering them increasingly susceptible to exploitation.

The application of deepfake audio can be categorized into three primary types: voice conversion, speech synthesis, and replay attacks. Replay attacks [1], [2] involve recording an

individual's voice and subsequently replaying it to bypass authentication mechanisms, posing a substantial threat to voice-based security systems. Speech synthesis [3], [4], on the other hand, employs AI to generate speech that closely mimics real voices, making it a powerful tool for fraudulent activities such as telecom scams, identity fraud, and misinformation dissemination. Similarly, voice conversion modifies existing speech recordings to resemble another person's voice, enabling malicious actors to impersonate individuals, fabricate false evidence, or manipulate digital communications.

With rapid advancements in AI-driven voice generation, particularly through generative adversarial networks (GANs) [5], deepfake technology has become increasingly

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sophisticated, posing significant risks to speaker recognition systems, including those used for one-time password verification. The exploitation of deepfake audio has raised serious concerns about financial security, as it has been leveraged for banking fraud, social engineering attacks targeting customer service representatives, and deceptive phone-based impersonation schemes.

The detection of synthetic speech manipulations has relied significantly on sophisticated approaches from both transfer learning and advanced machine intelligence, considering the growing complexity of artificially generated voice data [6]. Transfer learning [7] played a crucial role, as it enabled the utilization of pre-acquired representations from extensive corpora to enhance performance on novel and limited-scale datasets. This strategy significantly boosted recognition accuracy, particularly in scenarios where instances of synthetic voice manipulations were scarce.

The large-scale misuse of deepfake audio, particularly in fabricating voices of public figures, poses a substantial threat by propagating misinformation and eroding public trust. To counteract these threats, researchers have focused on real-time monitoring, feature extraction, and advanced machine learning (ML) algorithms for deepfake detection. Compared to conventional spectral analysis techniques, deep learning (DL) models leveraging frequency imperfection profiling exhibit superior performance in detecting synthetic audio. Additionally, multi-factor authentication methods [8] enhance the robustness of voice authentication systems by addressing inherent security vulnerabilities. Given the rapid evolution of deepfake technology, governments, organizations, and research institutions must enforce stringent regulations and invest in the development of more effective detection mechanisms to safeguard electronic communication.

In this study, we proposed robust advanced DL models to enhance generalization in deepfake audio detection. We systematically explore state-of-the-art methodologies for identifying deepfake speech, emphasizing the efficacy of various DL-based approaches as well as different feature combinations. Specifically, we conducted a comparative analysis of multiple models: BLSTM, custom CNN, ResCNN-Attention-BGRU, WIREnet, ResNet FC, and SE-Enhanced 1D-CNN by providing them with different feature combinations.

This manuscript outlines its main contributions in the subsequent points:

- We extract diverse audio features from both real speeches and deepfake speeches, including spectral centroid (SC), spectral rolloff (SR), chroma short-time energy transformation (chroma-STET), root mean square energy (RMSE), zero-crossing rate (ZCR), spectral bandwidth (SB), and mel-frequency cepstral coefficients (MFCCs). MFCCs, spectral, and temporal features play a significant role in distinguishing deepfake audio signals.

- We systematically compare the performance of multiple advanced DL architectures, BLSTM, custom CNN, ResCNN-Attention-BGRU, WIREnet, ResNet FC, and SE-Enhanced 1D-CNN, providing detailed insights into their effectiveness.
- We also explored various combinations of hybrid features and evaluated their effectiveness in distinguishing real from fake speech using advanced DL techniques.
- Our experimental results demonstrate that SE-Enhanced 1D-CNN achieves the highest accuracy of 97.64% in distinguishing the real and fake speech signals, surpassing all other models and highlighting its superior effectiveness in detecting deepfake audio.

This manuscript progresses as follows: Section II delivers an in-depth examination of the literature review. Section III outlines the methodological approach adopted in this study. Section IV introduces the proposed experimental setup developed for this study. In Section V, experimental findings are thoroughly evaluated, and a comparative assessment of the training models is conducted. Lastly, Section VI encapsulates the primary outcomes.

II. RELATED STUDIES

Reports on deepfake audio detection emphasize the inadequacy of evaluation and testing in existing studies, underscoring the need for more rigorous benchmarking methodologies. The effectiveness of ML models in this domain is fundamentally tied to the refinement of feature representations within classifiers. DL techniques [9], [10] have demonstrated superior efficacy in feature extraction and classification; however, they require substantial computational resources and extended processing time. Various transfer learning-based DL approaches have been explored to mitigate these challenges, addressing feature complexity and enhancing deepfake detection performance. These advancements highlight the urgent need for more sophisticated and robust solutions to counter the growing threat of audio deepfakes across multiple sectors.

Over the past decade, AI and ML techniques have significantly advanced audio event detection, yielding notable improvements in accuracy and efficiency [11]. However, the performance of the models is adversely affected by variations in speaking style, intonation, pronunciation, accent, and the speaker's dialect [12]. The study [13] introduced a feature reconstruction approach for anomaly detection in subsampled signals using an LSTM-based network. Unlike conventional methods, the proposed model directly processes subsampled time-domain signals to reconstruct the original feature vectors. Experimental results validate its effectiveness in identifying anomalies from subsampled signals. The authors [14] proposed an approach that integrates deep residual learning, specifically an adaptation of the ResNet-18 architecture. The experiments were done on the ASVspoof 2017 dataset, and the architecture attained 0% of an equal error rate (EER) on the evaluation set, surpassing

prior benchmarks of 0.12% and 2.76% respectively. These findings highlight the efficacy of transfer learning and feature representation in addressing the challenges of replay attack detection.

The study done by [15] presented a comprehensive database, evaluation protocols, and key experimental findings. The assessment was conducted under diverse acoustic recording and replay conditions, leading to a significant increase in the EER of a baseline automatic speaker verification (ASV) system from 1.76% to 31.46%. Among the 49 participating research teams, 20 achieved improvements over the baseline replay spoofing detector EER of 24.77%. Despite these advancements, the results highlight the persistent challenge of developing countermeasures robust to varying replay attack scenarios. To address these issues, [16] proposed a spoofing detection system integrating deep layer aggregation (DLA) and SincNet to enhance synthetic audio discrimination. DLA, a fully convolutional tree-structured model, aggregates multi-scale features to capture local and global speech patterns, while SincNet, as the frontend, directly derives fine-grained representations from raw audio for end-to-end learning. Experiments performed on the ASVspoof dataset demonstrate superior performance, with our single model achieving an EER of 13.99%. Furthermore, [10] developed a cross-domain approach that integrates spatial and frequency-aware cues for deepfake detection. An adaptive fusion module integrates learned features with RGB cues, boosting classification accuracy. Experiments on benchmark datasets, including FaceForensics++ (FF++), confirm the method's robustness, achieving over 98% accuracy across varied deepfake techniques.

These existing studies explored various methodologies for deepfake audio detection, encompassing ML models, feature extraction techniques, and datasets employed in existing research. Advanced DL-based approaches have been developed to enhance detection capabilities. For example, Kanwal et al. [1] introduced a model incorporating VGG with convolutional block attention mechanisms, significantly improving classifier performance in fake speech detection and achieving 99.40% on the ASVspoof 2019 corpus. Deepfake audio detection has garnered significant attention, with numerous studies leveraging GAN-specific artifacts to develop robust detection models [5], [8]. Since GANs are widely used for synthetic speech generation, their artifacts serve as crucial indicators for distinguishing between authentic and AI-generated voices. One of the most effective deepfake detection techniques involves spectrogram analysis, as synthetic speech often exhibits distinctive distortions observable in spectral representations of audio signals [3]. Additionally, transformer-based architectures, such as Wav2Vec and Whisper, have been extensively explored as discriminators capable of differentiating real and synthetic voices [17]. The growing prevalence of synthetic speech in malicious activities, including identity theft, disinformation campaigns, and fraudulent impersonation, necessitates a renewed focus

on deepfake audio detection. Recent advancements in DL have enabled the synthesis of highly realistic voice generation and conversion, complicating detection efforts.

The study [3] proposed a hybrid CNN-LSTM framework to enhance synthetic speech detection. Experimental results demonstrated superior performance across multiple datasets, including ASVspoof2019, FakeAVCeleb, and in-the-wild recordings, highlighting the robustness of the proposed approach. The hybrid architecture significantly improved the temporal modeling of speech signals, a critical factor in deepfake speech detection. The extraction of features has a significant influence on enhancing the model performance, with prior research exploring various spectral and temporal features. For instance, [1] employed MFCCs combined with self-attention mechanisms for replay attack detection, achieving state-of-the-art results on the ASVspoof 2021 dataset. Similarly, [18] investigated the effectiveness of chroma-based spectral analysis for identifying inconsistencies in synthesized speech, demonstrating that integrating chroma features with conventional spectrogram-based approaches led to significant improvements in classification accuracy. Reliable benchmarking of deepfake detection models necessitates high-quality datasets. Several researchers have introduced diverse datasets to facilitate fair evaluations [19]. Notably, [8] presented the ASVspoof 2019 corpus, which includes the logical access and physical access attack scenarios, making it a widely adopted benchmark in the field. Many detection models have leveraged the ASVspoof2019 dataset for performance evaluation [10], [20].

In [21], two distinct strategies were explored: feature extraction-based and image representation-based. The former transformed audio signals into spectral descriptors, which were then analyzed using classical classification models. The latter encoded the signals into mel-spectrograms, processed via a spatial transformer network (STN) and temporal convolutional network (TCN). TCN was selected due to its effectiveness in handling temporal dependencies in sequence data. Experimental findings indicated that deep architectures, especially TCN, surpassed traditional methods, achieving a peak accuracy of 92%. Kilinc et al. [22] utilized MFCCs as input features for both ML and DL classifiers to determine the authenticity of audio samples, distinguishing between real and fake speeches. The authors [23] proposed a framework titled MFCC-GNB XtractNet to address the challenge of synthetic voice detection. The process involved deriving MFCC from speech signals to capture critical auditory features. Comparative evaluation using architectures such as logistic regression, RF, and K-nearest neighbors was conducted through k-fold cross-validation. The proposed system yielded superior performance, attaining a peak accuracy of 99.93%. In [24] initiated with assembling a balanced speech dataset comprising real and artificially generated audio across various linguistic and emotional categories. Audio inputs were standardized and denoised, followed by the extraction of meaningful acoustic representations. A CNN

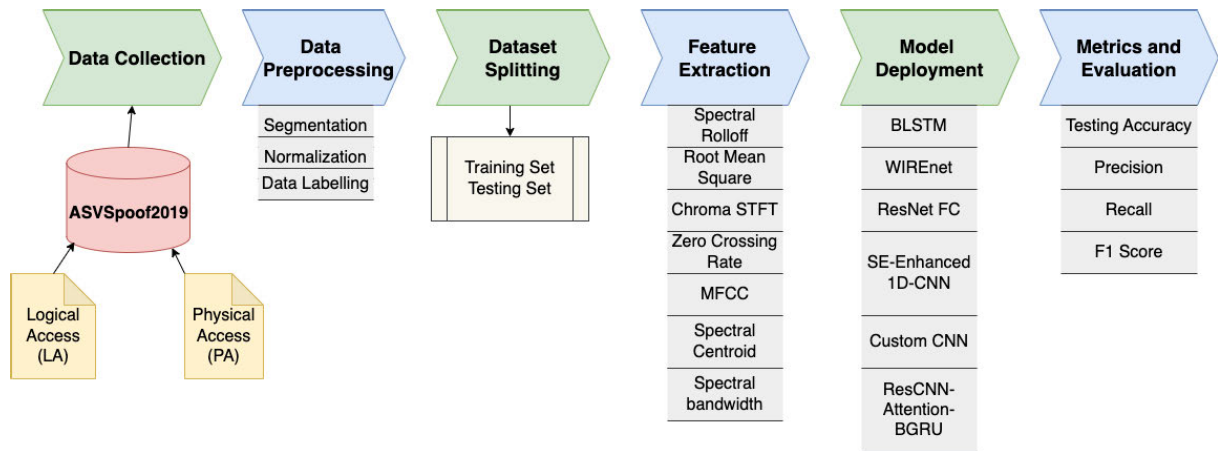


FIGURE 1. Workflow of proposed approach for detection of deepfake audios.

combined with recurrent components was developed to capture spectral-temporal dynamics from spectrogram inputs. This study [25] introduced Arabic-AD for detecting audio deepfakes through self-supervised learning (SSL) techniques, which identified both fake and real speeches. A series of three comprehensive experiments was conducted to evaluate and compare the proposed method with established benchmarks. The results demonstrated that Arabic-AD surpassed other leading techniques, achieving the lowest 0.027% of EER and 97% of accuracy in distinguishing real and fake voices.

Thus, neural network-based anomaly detection models, designed to capture inconsistencies in speech patterns, provide a promising approach for detecting AI-generated audio. Complementary to these methods, advanced audio feature extraction techniques, including MFCCs and chroma features, have been widely integrated into deepfake detection frameworks to enhance performance and reliability. The literature underscores the critical need for robust and efficient deepfake detection systems, particularly in response to vulnerabilities introduced by advanced speech synthesis techniques such as voice conversion, text-to-speech, and replay attacks. Also, the rapidly evolving landscape of synthetic speech generation, where no single architecture has been conclusively established as optimal for detecting audio forgeries across varying conditions [26]. In this study, we have extracted both temporal and spectral features from the speech signals, and then a comparative in-depth analysis is done to evaluate the performance of advanced ML and DL models in distinguishing real and synthetic audio, aiming to identify the most accurate and effective approach.

III. METHODS AND MATERIALS

This section outlines the adopted research approach, detailing the processes of acquiring data, performing initial preprocessing, extracting relevant features, and constructing the ML and DL framework. A visual representation of the complete proposed strategy is illustrated in Figure 1.

A. DATA COLLECTION

High-quality datasets are essential for training deepfake detection models, as they provide the diversity and robustness required to develop effective solutions. The widely recognized ASVspoo 2019 dataset provides a broad spectrum of spoofing techniques, including replay attacks, voice conversion, and text-to-speech. The ASVspoo 2019 competition was supported by leading researchers who sought to push the boundaries of ASV technology by introducing a benchmark dataset that systematically integrated various spoofing attacks. A key advancement in ASVspoo 2019 [16] was the integration of TTS and voice conversion systems, an initiative that extended prior challenges and allowed researchers to assess vulnerabilities introduced by increasingly sophisticated spoofing techniques.

In our methodology, we utilized the ASVspoo 2019 [27] dataset for experimentation, encompassing a well-balanced collection of real as well as fake speeches. In the dataset, the fake audio utterances are generated using diverse algorithms with differing acoustic characteristics, ensuring significant variety in spoofing techniques. The dataset includes multiple English-native speakers with neutral reading style, and it covers variability in speaker identity, but does not explicitly give information about a broad range of global accents or expressive speaking styles. After applying down-sampling techniques to ensure an equal distribution of classes, the dataset consisted of 12,493 real and 12,493 fake audio samples. To train and evaluate our models effectively, we performed an 80:20 train-test split. Consequently, the training set contained 9,994 real and 9,994 fake samples, while the test set included 2,499 real and 2,499 fake samples. The distribution of samples across these subsets is illustrated in Figure 2. This balanced distribution ensures that proposed models do not develop a bias toward either class, thereby enhancing their robustness in distinguishing between natural and synthetic speech signals.

The experiments were conducted on the English-language ASVspoof 2019 dataset, used in spoofing countermeasure research in ASV. English serves as a suitable baseline due to its extensive availability of annotated corpora. However, language dependency arises due to differences in phoneme inventories, phonotactic rules, prosodic patterns, and speech production characteristics [28]. These variations influence acoustic and spectral features, potentially affecting both feature extraction methods (e.g., MFCC) and the learned representations in deep learning models.

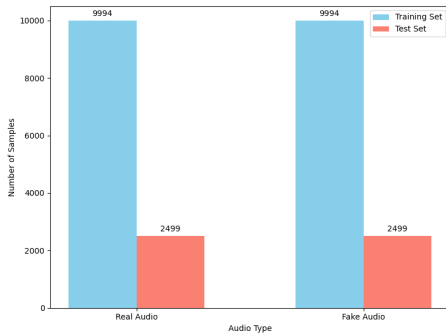


FIGURE 2. Distribution of real and fake audio samples in training and test sets in the AVSspoof2019.

B. DATA PREPROCESSING

It is a fundamental step in DL, as it directly impacts the performance and generalization of ML and DL models. This process encompasses multiple stages, including data cleaning, transformation, and dataset construction [18]. In our proposed work, we implemented various preprocessing approaches, like normalization, speech segmentation, and data labeling, to ensure high-quality inputs for the proposed models. Specifically, real speech samples were assigned a label of 1, whereas synthetic or altered audio data were labeled as 0 to distinguish between real and fake speech samples effectively.

C. FEATURE EXTRACTION

The synthetic audio generated through deepfake techniques frequently exhibited acoustic patterns analogous to those found in authentic speech. Nevertheless, enhancing the ability of DL frameworks to differentiate between genuine and manipulated signals remained a significant challenge. Consequently, the type and quality of the extracted acoustic representations substantially influenced the performance and classification reliability of the predictive model. Various authors have employed different types of extracted features, which are subsequently fed into DL-based frameworks for detection. For instance, [21] utilized both spectral features and image-based representations to effectively determine the authenticity of the given audio samples. In another study, [22] employed MFCC as the primary feature representation, which was subsequently used as input to models for the detection of natural and artificially generated audio speeches.

Similarly, in the study conducted by [23], MFCC features were employed for the detection of Non-spoofed and spoofed audio speeches. The authors further performed a comparative assessment by evaluating the efficacy of various ML and DL models on the extracted features.

In this study, we implemented a systematic feature extraction pipeline to analyze deepfake and real audio samples for fake audio detection. To ensure uniformity across all samples, each audio file was preprocessed to a fixed duration of four seconds through either truncation or zero-padding. This standardization minimized variability in input length and ensured consistency in feature extraction. A diverse set of acoustic features was extracted, including chroma-STFT, RMSE, SC, SB, SR, and ZCR. Additionally, 13, 26, and 39 MFCCs were computed to capture essential speech characteristics. All the extracted features were then combined and used as input to the model. These features were selected for their effectiveness in distinguishing real and synthetic speech signals based on spectral and temporal properties.

All extracted features of the speech signals were normalized using normalization to guarantee that every speech in the dataset had a mean-centered and variance-scaled representation, thereby enhancing the training process and reducing computation. By reducing the possibility of biases brought on by differing feature value scales, this normalization improves the robustness and convergence of DL models. The dataset, comprising real and fake audio samples stored in FLAC format, was processed in a structured manner by iterating through labeled directories. The extracted feature set was then stored in a CSV file for further analysis. This preprocessed dataset forms the basis for training robust DL models for deepfake speech identification, ensuring reliable classification and generalization across diverse audio samples.

The audio signal features used in this study are briefly discussed below:

- 1) **Spectral Rolloff (SR):** It measures the frequency at which a certain percentage (generally 85 to 95 percent) of the overall spectral energy is focused. Necessarily, it helps determine the single highest point of the frequency range in which an audio signal has the most energy. The mathematical expression is given in equation 1

$$\sum_{m=0}^{m_r} |X(m)| = p \sum_{km=0}^{N-1} |X(m)| \quad (1)$$

where:

- $X(m)$ represents the magnitude of the Fourier Transform at frequency bin m .
 - m_r represent rolloff frequency bin.
 - p is the percentage (e.g., 0.85 or 0.95).
 - N represents the total number of frequency bins.
- 2) **Zero Crossing Rate (ZCR):** It is a fundamental audio feature that measures how frequently the signal changes its sign (crosses zero) within a given time

frame. For a discrete-time signal $x[n]$, ZCR is computed as:

$$ZCR = \frac{1}{L-1} \sum_{n=1}^{L-1} \mathbb{I}(x(n) \cdot x(n-1) < 0) \quad (2)$$

where:

- $x(n)$ represents the discrete-time signal.
- L is the total number of samples in the frame.
- $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the argument is true, otherwise 0.
- $x(n) \cdot x(n-1) < 0$ checks if a zero crossing occurs between two consecutive samples.

- 3) **Root Mean Square Energy (RMSE):** It is a measurement of the average power or loudness of an audio signal, which impacts the capturing intensity and dynamics of the signal. A highly fluctuating energy signal is typically characteristic of a real audio phenomenon. Given a discrete audio signal $x[n]$, the RMS energy over a window of length N is

$$E_{RMSE} = \sqrt{\frac{1}{M} \sum_{n=0}^{M-1} x^2(n)} \quad (3)$$

where:

- $x(n)$ represents the discrete-time signal.
- M represents the total number of samples.

- 4) **Chroma short-time energy transformation (chroma-STFT):** It is an audio feature that, through the mapping of frequency components into a total of 12 chroma (pitch class) bins, gives a representation of the harmonic and melodic content of a signal. These bins correspond to the twelve notes in an octave (C, D, etc.). To compute CSTFT, the short-time Fourier transform is first computed and then converted into a chroma representation by grouping frequency bins into 12 different pitch classes (chroma bins).
- 5) **Spectral Centroid (SC):** It gives the “center of mass” of the power spectrum. It indicates where most of the energy in the frequency domain is concentrated, often correlating with the perceived brightness of a sound.

$$SC = \frac{\sum_{k=0}^{M-1} f_k |X(k)|}{\sum_{k=0}^{M-1} |X(k)|} \quad (4)$$

where:

- f_k is the frequency corresponding to the k th bin.
- $X(k)$ represents the magnitude of the Fourier transform for each frequency bin k .
- M represents the complete set of frequency bins

- 6) **Spectral Bandwidth (SB):** It is used to measure the spread of frequencies around the SC. It quantifies how wide or narrow the frequency distribution is, often related to the richness or noisiness of a sound.

$$SBW = \sqrt{\frac{\sum_{k=0}^{M-1} |X(k)| (f_k - SC)^2}{\sum_{k=0}^{M-1} |X(k)|}} \quad (5)$$

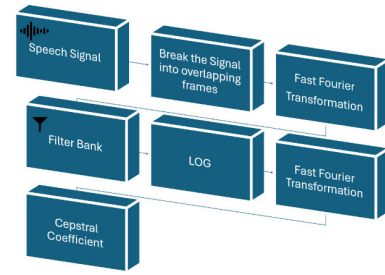


FIGURE 3. Steps for the extraction of MFCC from audio signals.

where:

- f_k is the frequency corresponding to the k th bin.
- $X(k)$ represents the magnitude of the Fourier transform for each frequency bin k .
- SC is the SC, given by:

$$SC = \frac{\sum_{k=0}^{M-1} f_k |X(k)|}{\sum_{k=0}^{M-1} |X(k)|}$$

- M represents the complete set of frequency bins

- 7) **Mel-Frequency Cepstral Coefficient:** It represents the short-term power spectrum of an audio signal on a non-linear Mel scale, derived from the logarithm of the power spectrum and decorrelated via the discrete cosine transform (DCT). They are widely used in speech and audio processing due to their effectiveness in capturing timbral features. MFCC13 primarily captures broad spectral features, which are useful for speech recognition but may miss. MFCC29 includes first-order derivatives Δ , improving sensitivity to temporal changes. MFCC39 adds second-order derivatives Δ^2 , which track more complex variations in speech dynamics, helping detect unnatural transitions often present in deepfake audio.

The MFCC extraction process is illustrated in Figure 3, and MFCCs for real and fake audio samples are shown in Figures 4 and 5, respectively. Due to their resemblance to the human auditory system, MFCCs are an effective feature representation for characterizing the spectral properties of audio signals response [29]. In real audio MFCC refer 4, red intensity bands appear more localized in time, steady harmonic structure, and stable formant positions, dB values range widely, but transitions between low and high energy regions are smoother. In contrast, the MFCC of a fake audio sample refers 5 red intensity bands are evenly spread overall, with some unexpected spikes, unnatural periodicity in horizontal patterns. The transitions between energy levels are more abrupt. Thus, the varying patterns present in the audio samples enable deep learning models to identify and determine whether a given sample is real or synthetic.

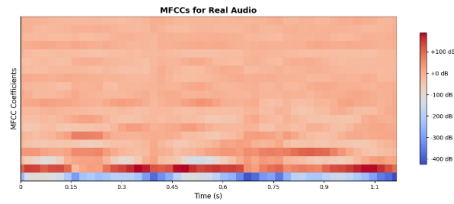


FIGURE 4. MFCC of the real audio signals.

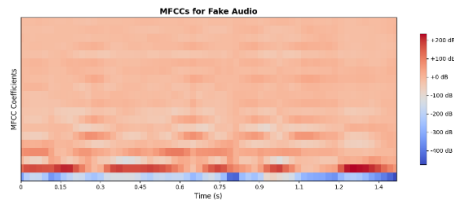


FIGURE 5. MFCC of the fake audio signals.

D. MODEL DEPLOYMENT

For model development, we explored DL architectures to effectively process extracted feature representations and classify audio signals as either genuine or artificially generated. This research primarily seeks to evaluate the capability of these architectures in distinguishing between real and fake speech, thereby assessing their robustness in authenticity detection. To achieve this, we systematically selected and configured different models, considering their structural composition and parameterization during the experimental phase. The subsequent sections provide a detailed overview of the chosen architectures and their formation throughout the experimental process.

- BLSTM:** This BLSTM-based neural network is designed for deepfake audio detection, leveraging bidirectional temporal dependencies. It begins with a BLSTM layer (128 units), followed by a dropout layer (0.3) to overcome overfitting. A second BLSTM layer contains 64 units with an additional dropout layer (0.3) that refines the learned patterns, while a final BLSTM layer has 32 units, extracts high-level features, before layer normalization is applied and stabilizes activations. Subsequently, two fully connected layers are applied, with the first containing 32 units and the second containing 14 units, both utilizing the activation function as ReLU. A dropout layer with a rate of 0.4 is employed to enhance feature extraction. The output layer, containing 1 unit and a sigmoid activation function, is applied to detect whether the audio speech is real or synthetic. The model is optimized using the Adam optimizer with 0.0001 as the learning rate with binary cross-entropy loss; the model incorporates a ReduceLROnPlateau [30] scheduler for an adaptive learning rate adjustment, ensuring stable training and improved generalization.
- WIREnet:** In the proposed study, WIRENet is a hybrid CNN-BLSTM model for deepfake audio detection. The architecture begins with stacked Conv1D layers ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$) with an activation function as ReLU, followed by batch normalization and max pooling to retrieve hierarchical spectral and temporal features. BLSTM layers with 128 units capture sequential dependencies, accompanied by a dropout layer configured with a rate of 0.5, to mitigate overfitting. A global average pooling layer reduces dimensionality, followed by a fully connected layer (256, ReLU, L2 regularization) for feature enhancement. The final Dense(1, sigmoid) layer outputs the classification probability. Trained with Adam (learning rate = 0.000006) and binary cross-entropy loss, the model integrates CNN's feature extraction and LSTM's sequential learning for robust deepfake detection.
- ResNet FC:** It is designed for deepfake audio detection, leveraging residual connections to enhance learning. The model begins with a fully connected layer containing 128 units, and the activation function is chosen as ReLU, followed by batch normalization for stable training. It incorporates two residual blocks, each containing two fully connected layers with batch normalization and a skip connection (add layer) to overcome the problem of the vanishing gradient. After feature extraction, the model utilizes a fully connected layer with 64 units followed by a ReLU activation function. Subsequently, another dense layer with 32 units and ReLU activation is employed to achieve further feature abstraction. Ultimately, the architecture concludes with an output fully connected layer with a sigmoid activation to facilitate detection into two categories. Model optimization is carried out utilizing the Adam algorithm, configured with a learning rate of 9×10^{-6} , across 50 training cycles and a mini-batch size set to 64.
- SE Enhanced 1D-CNN:** The model employs ResNet with squeeze-and-excitation attention for detecting real and fake audio. The first convolutional block comprises a Conv2D layer equipped with 64 kernels of dimensions (1, 3), incorporating L2 kernel regularization to mitigate overfitting, with batch normalization applied thereafter and Leaky ReLU activation for stable learning. Max-Pooling2D and a 0.3 dropout further enhance generalization. The first residual block integrates Conv2D layers with batch normalization and residual connections to prevent vanishing gradients. A second convolutional block refines features before passing them through an SE attention layer, which adaptively reweights feature maps to highlight deepfake-specific patterns. The classification stage includes fully connected layers with a 0.4 dropout for regularization, concluding with a softmax output layer. The architecture is trained utilizing focal loss to emphasize hard-to-classify examples and optimized with Adam, alongside a cosine learning rate scheduler for efficient convergence.

- **Custom CNN:** This custom CNN model is designed for deepfake audio detection. It begins with a convolutional layer comprising 32 filters of size 2×2 , utilizing ReLU activation and L2 regularization to mitigate overfitting. A max-pooling layer follows, reducing the height dimension while preserving spatial information along the width. The second convolutional layer, containing 64 filters and having a kernel size 2×1 , is also succeeded by max-pooling. The output is then flattened and passed through a fully connected layer with 128 ReLU-activated units, followed by a dropout for regularization. Finally, a softmax-activated dense layer performs multi-class classification. The network is optimized using the Adam optimizer with categorical cross-entropy loss and an exponential learning rate schedule, ensuring effective training in mutually exclusive classification scenarios.
- **Residual CNN + self-attention + BGRU:** It begins with a convolutional block containing 64 filters (kernel size = 5), with batch normalization applied thereafter, swish activation, and then a dropout layer is applied for regularization. A residual block enhances feature extraction through residual connections and a similar convolutional structure. A second convolutional block with 128 filters extracts higher-level features, while a self-attention mechanism emphasizes critical aspects of the audio sequence. A BGRU layer captures sequential dependencies, followed by global average pooling to condense features into a fixed-size vector. This vector is processed by two dense layers with Swish activation and dropout, ensuring effective regularization. The final softmax layer performs classification. The model is optimized using AdamW with custom focal loss to handle class imbalance, while a learning rate scheduler and early stopping improve training efficiency.

E. METRICS EVALUATION

The performance of a system serves a critical role in ensuring the credibility of classification and recognition tasks, particularly when conducting comparative analyses. To evaluate the effectiveness and robustness of the developed model, we employ four fundamental mathematical metrics: accuracy, recall, precision, and F1-score. Mathematical formulations of these evaluation metrics are presented below.

$$\text{Accuracy} = \frac{T_{pos} + T_{neg}}{T_{pos} + T_{neg} + F_{pos} + F_{neg}} \quad (6)$$

$$\text{Precision} = \frac{T_{pos}}{T_{pos} + F_{pos}} \quad (7)$$

$$\text{Recall} = \frac{T_{pos}}{T_{pos} + F_{neg}} \quad (8)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

where

T_{pos} , F_{pos} , T_{neg} , F_{pos} , F_{neg} are the true positive, false positive, true negative, and false negative, respectively.

IV. PROPOSED FRAMEWORK

Initially, each audio sample undergoes preprocessing, during which speech labeling is performed to differentiate between real and spoofed speech samples. After preprocessing, the labeled speech signals proceed to the feature extraction phase, where both spectral and temporal features of the signals are extracted to capture meaningful information from the speech data.

Next, temporal features are extracted to capture the essential characteristics of the speech waveform, which are crucial for effective classification. We utilized three sets of features extracted from the speech signals to determine which representation provides the best performance in distinguishing between real and fake speech. The three feature representations are presented in Table 1. To ensure a balanced dataset, we applied a downsampling strategy that equalizes the number of bona fide and spoofed speech samples, preventing model bias towards the majority class. The balanced dataset is then partitioned into a training set and a testing set using an 80-20 split, keeping the class balance intact to ensure fair evaluation.

After the data splitting, we proposed a robust ML and DL framework for the speech is real or fake. The dataset is first preprocessed by extracting both temporal and spectral features from speech signals, which are then combined to form a feature representation. The speech signal then undergoes a preprocessing stage, where data labeling is performed to distinguish between bona fide and spoofed speech samples. Following pre-processing, the labeled speech signals are forwarded to the feature extraction phase.

For the experimental evaluation, we conducted the experiments using Google Colab, leveraging the NVIDIA T4 GPU to accelerate computations and efficiently process large-scale speech data. The high-performance GPU facilitated faster model training and evaluation, ensuring optimized execution of deep learning algorithms. The proposed framework adopted in this study is illustrated in Figure 6 below.

TABLE 1. The feature sets extracted from the real and fake audio signals.

Type of Features	Feature	Feature Set-01	Feature Set-02	Feature Set-03
Time-domain Features	ZCR	1	1	1
	RMS Value	1	1	1
	MFCC	13	26	39
Spectral Features	Chroma	1	1	1
	STFT	1	1	1
	Spectral Centroid	1	1	1
	Spectral Rolloff	1	1	1
	Spectral Bandwidth	1	1	1
Total Features		19	32	45

V. RESULTS AND DISCUSSIONS

This section presents a comparative analysis of DL architectures for deepfake audio detection. The evaluated models include BLSTM, WIREnet, ResNet FC, Residual CNN with Attention, and a custom CNN model. Model performance is

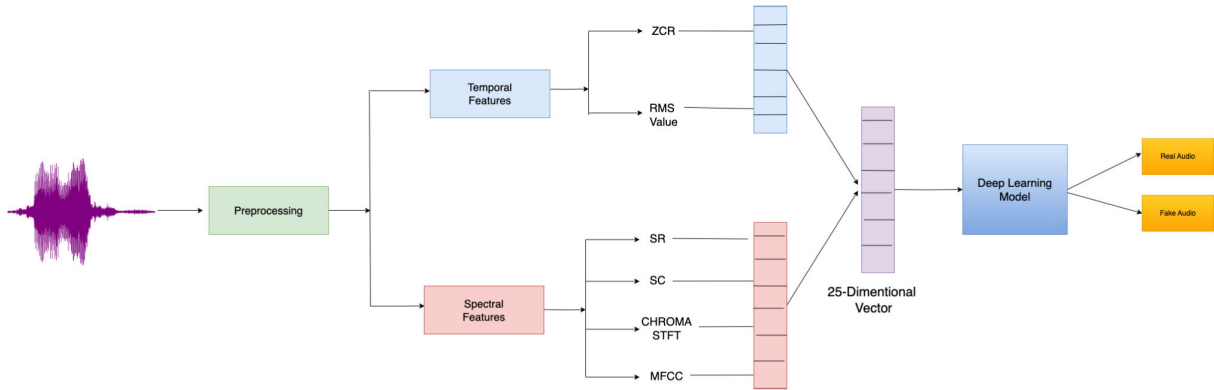


FIGURE 6. The overview of the proposed framework used in this study.

assessed based on four essential metrics: accuracy, precision, recall, and F1-score. Experiments were conducted using all three sets of feature representations, and the results are analyzed and discussed in this section.

Figures 7 and 8 illustrate the training accuracy and loss, respectively, during the training of advanced models such as BLSTM, WIREnet, ResNet FC, Residual CNN with Attention, and a custom CNN model. As observed, increasing the epochs improves the training accuracy while decreasing the training loss, indicating that the models are effectively learning from the dataset and capturing patterns that distinguish real and fake speech samples. Also, Figure 9 presents the training and testing accuracies of the proposed models. This figure demonstrates that there is no evidence of overfitting, and the proposed models perform well on unseen test data, thereby confirming their generalizability.

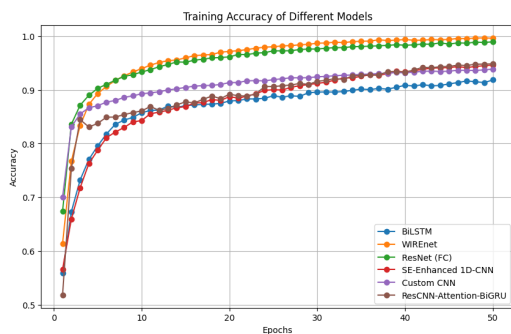


FIGURE 7. Comparison between the training accuracy of different models.

A. RESULTS AND DISCUSSION ON FEATURE SET-01

Table 2 summarizes the comparative performance of different DL models on Feature Set-01. The findings illustrate that the ResCNN-Attention-BGRU model demonstrates the highest accuracy among all evaluated models, achieving the highest accuracy, recall, precision, and F1-score of 0.93. ResNet FC and WIREnet also demonstrate strong performance, with 0.9225 and 0.9147 accuracy, indicating the advantage of deep residual connections and specialized architectures



FIGURE 8. Comparison of training loss by different models.

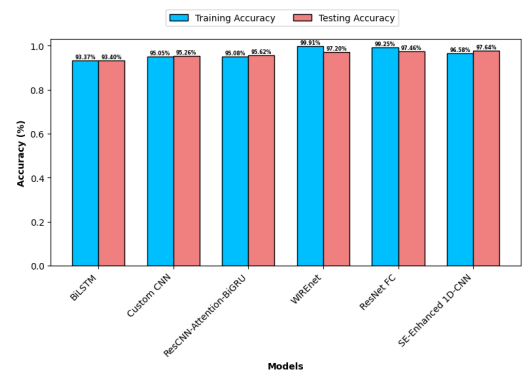


FIGURE 9. Training and testing accuracies in the Deepfake detection experiments.

in capturing complex speech patterns. The other models achieved comparable performance, with the custom CNN attaining an accuracy of 0.8401, the SE-Enhanced 1D-CNN reaching 0.8304, and the BLSTM model achieving 0.8246. These results illustrate that models incorporating deeper feature extraction, residual learning, and attention mechanisms are more effective for deepfake audio detection compared to basic CNN or LSTM-based architectures. Additionally, Figure 10 presents the confusion matrices, and Figure 11 presents the ROC curves of the proposed models

TABLE 2. Model performance comparison for the feature Set-01.

Model	Accuracy	Precision	Recall	F1-Score
BLSTM	0.82	0.84	0.82	0.82
SE-Enhanced 1D-CNN	0.83	0.85	0.83	0.83
Custom CNN	0.84	0.84	0.84	0.84
WIREnet	0.91	0.92	0.91	0.91
ResNet FC	0.92	0.92	0.92	0.92
ResCNN-Attention-BGRU	0.93	0.93	0.93	0.93

evaluated on the testing set, illustrating the performance of each model on the test data.

B. RESULTS AND DISCUSSION ON FEATURE SET-02

Table 3 illustrates a comparative evaluation of various models based on their precision, accuracy, recall, and F1-score using Feature Set-02. Among the models, the BLSTM obtained an accuracy of 93.16%. The custom CNN model further improved the results, achieving an accuracy of 94.46%, indicating the effectiveness of convolutional architectures in capturing relevant features. Notably, the ResCNN-Attention-BGRU model exhibited a significant performance boost, reaching 95.96% accuracy, likely due to the combined benefits of residual connections, attention mechanisms, and bidirectional recurrent units. The SE-Enhanced 1D-CNN model, WIREnet, and ResNet FC models delivered the highest performances, with accuracies of 97.18%, 97.28%, and 97.42%, respectively. These results suggest that models incorporating advanced feature recalibration mechanisms (such as Squeeze-and-Excitation blocks) and deep residual architectures are particularly effective for time series classification tasks, achieving near-perfect precision, recall, and F1-scores across the board. The confusion matrices of the proposed models, evaluated on the testing set, are presented in Figure 12, illustrating how well each model performed on the test data. Furthermore, the ROC curves of all the models are shown in Figure 13.

TABLE 3. Model performance comparison for the feature Set-02.

Model	Accuracy	Precision	Recall	F1-Score
BLSTM	0.93	0.94	0.93	0.93
Custom CNN	0.94	0.94	0.94	0.94
ResCNN-Attention-BGRU	0.96	0.96	0.96	0.96
SE-Enhanced 1D-CNN	0.97	0.97	0.97	0.97
WIREnet	0.97	0.97	0.97	0.97
ResNet FC	0.97	0.97	0.97	0.97

C. RESULTS AND DISCUSSION ON FEATURE SET-03

Table 4 provides a comprehensive comparison of model performances for the Feature Set-03 using the evaluation metrics accuracy, precision, recall, and F1-score. Traditional ML models, such as DL models like BLSTM, demonstrated comparable performance, each achieving 93% accuracy, with consistent recall, precision, and F1-score values. The custom CNN architecture showed a modest improvement,

achieving 95% across all metrics, indicating its superior ability to capture relevant features. Further enhancements were observed with the ResCNN-Attention-BGRU model, which reached 96% in all performance indicators, reflecting the effectiveness of incorporating attention mechanisms and recurrent units. Notably, WIREnet and ResNet FC models both achieved 97% performance across metrics, suggesting that deeper architectures with optimized residual connections significantly contribute to better generalization on the test set. The highest performance was recorded by the SE-Enhanced 1D-CNN architecture, attaining 97.64% accuracy and achieving 98% in recall, precision, and F1-score, underscoring the efficacy of integrating squeeze-and-excitation blocks in enhancing feature representations for improved classification outcomes. The confusion matrices of the developed models, evaluated on the testing set, are shown in Figure 14, illustrating how well each model performed on the test data. Similarly, the ROC curves of all the models are presented in Figure 15.

TABLE 4. Model performance comparison for the feature Set-03.

Model	Accuracy	Precision	Recall	F1-Score
BiLSTM	0.93	0.94	0.93	0.93
Custom CNN	0.95	0.95	0.95	0.95
ResCNN-Attention-BGRU	0.96	0.96	0.96	0.96
WIREnet	0.97	0.97	0.97	0.97
ResNet FC	0.97	0.97	0.97	0.97
SE-Enhanced 1D-CNN	0.98	0.98	0.98	0.98

D. PERFORMANCE WITH MFCC VS MFCC + EXTRA FEATURES

To assess the impact of incorporating temporal and spectral features beyond MFCC, experiments were conducted using two configurations: (i) models trained exclusively on MFCC features, and (ii) models trained on MFCC combined with additional temporal and spectral features. As shown in Table 5, integrating these features improved the performance of all proposed DL architectures across accuracy, precision, recall, and F1-score metrics. This performance reveals that using both temporal and spectral information enables the proposed models to capture richer representations of the audio signal, thereby enhancing the accuracy for detecting the real and fake samples.

E. ABLATION STUDY

To check the impact of the components of the SE-enhanced 1D CNN architecture, we have performed a detailed ablation study, and the results reported are presented in Table 6, effectively quantifying the performance impact of including residual connections, squeeze-and-excitation (SE) attention, batch normalization, dropout, and L2 regularization.

In the ablation study, the shallow model exhibited a substantial reduction in accuracy (−2.74%). The removal of batch normalization resulted in a notable 1.61% performance drop, reinforcing its well-established role in accelerating

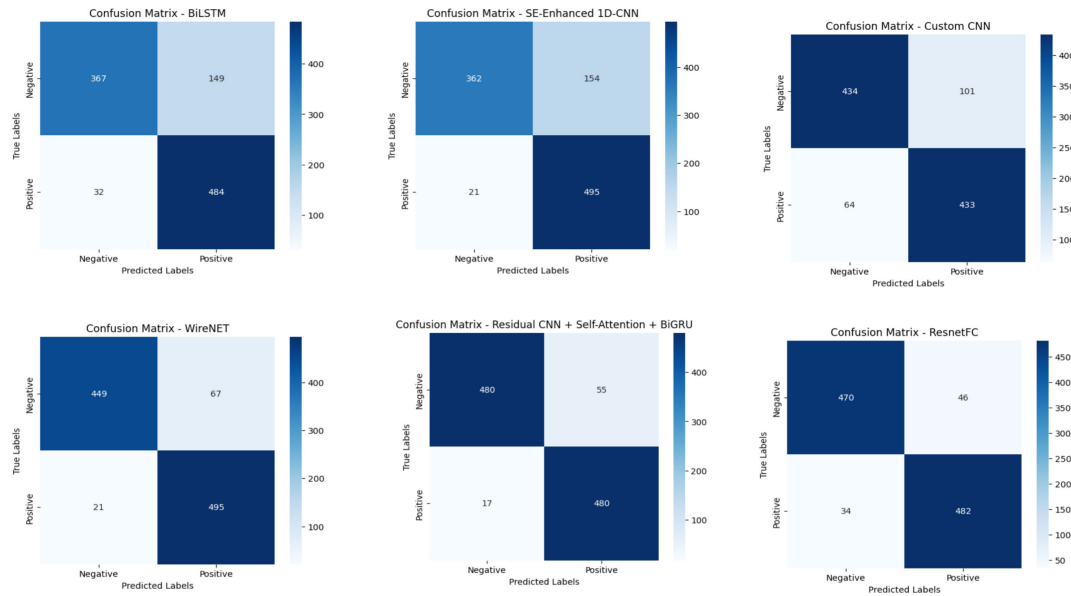


FIGURE 10. Confusion matrices of the proposed models for the feature Set-01.

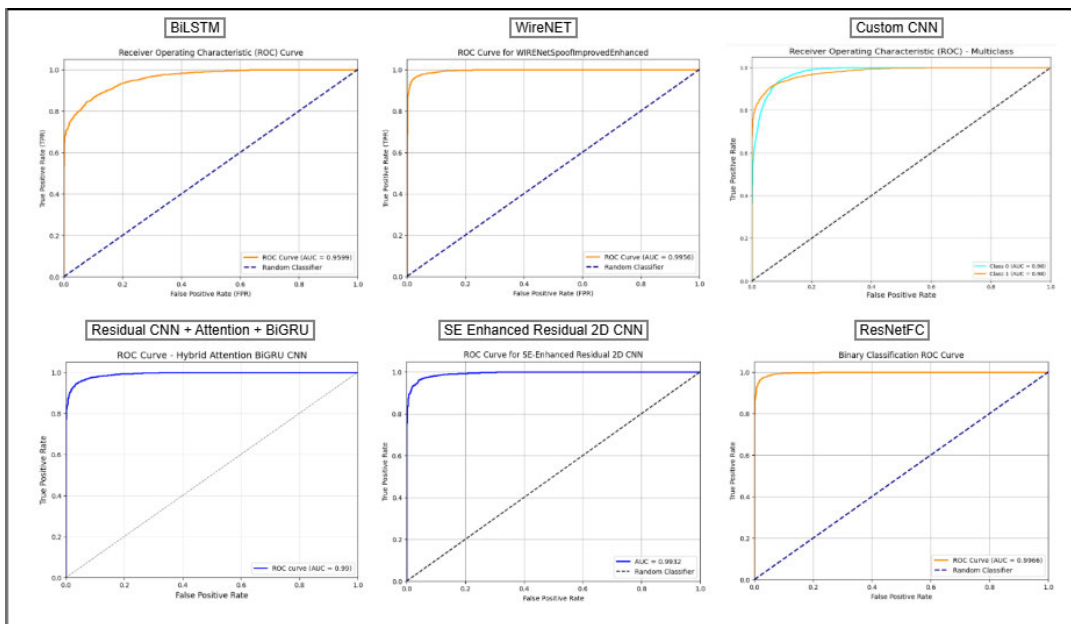


FIGURE 11. ROC curves of the proposed models for the feature Set-01.

training. Performance reductions observed upon the removal of SE attention (-1.17%) and residual connections (-1.08%) further highlight their meaningful contributions to the architecture, whereas the smaller 0.36% drop from removing L2 regularization indicates a comparatively minor effect. Replacing focal loss with a standard loss function had only a minimal impact (-0.10%), potentially suggesting that class imbalance is adequately addressed. Interestingly, the exclusion of dropout led to an improvement of approximately 1.43% in accuracy, revealing that

the model is not overfitting in the current experimental setup.

F. COMPARISON WITH EXISTING MODELS

Table 7 presents a comparative analysis of different models evaluated on the ASVspoof 2019 dataset. Traditional approaches such as SVM achieved accuracies ranging from 89% to 92.89% , while deep learning-based models like CNN+BiLSTM and Xception demonstrated stronger performance, reaching 97.21% and 92.11% , respectively.

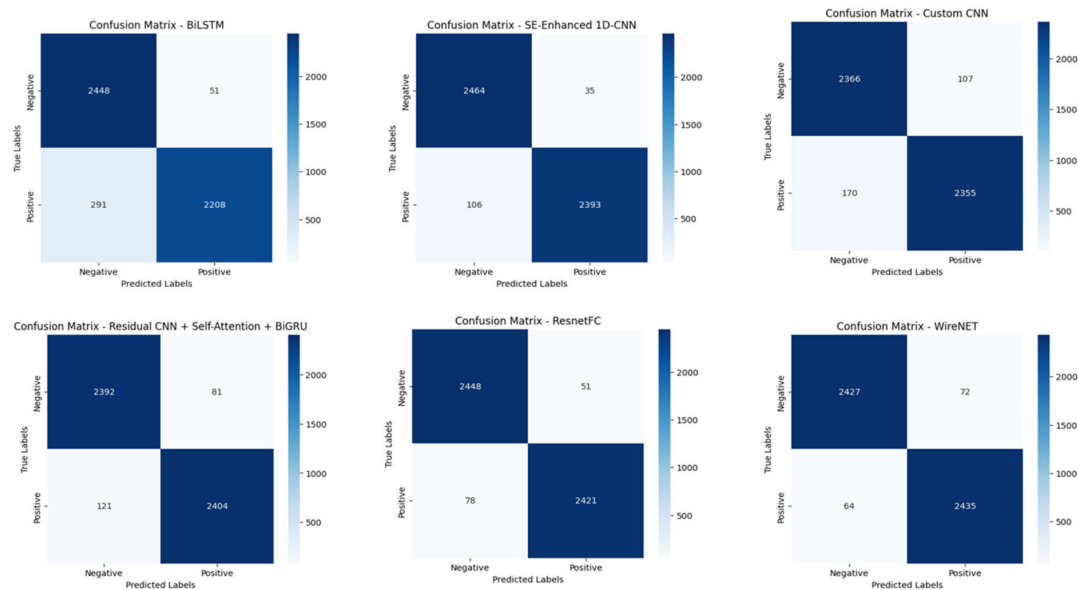


FIGURE 12. Confusion matrices of the proposed models for the feature Set-02.

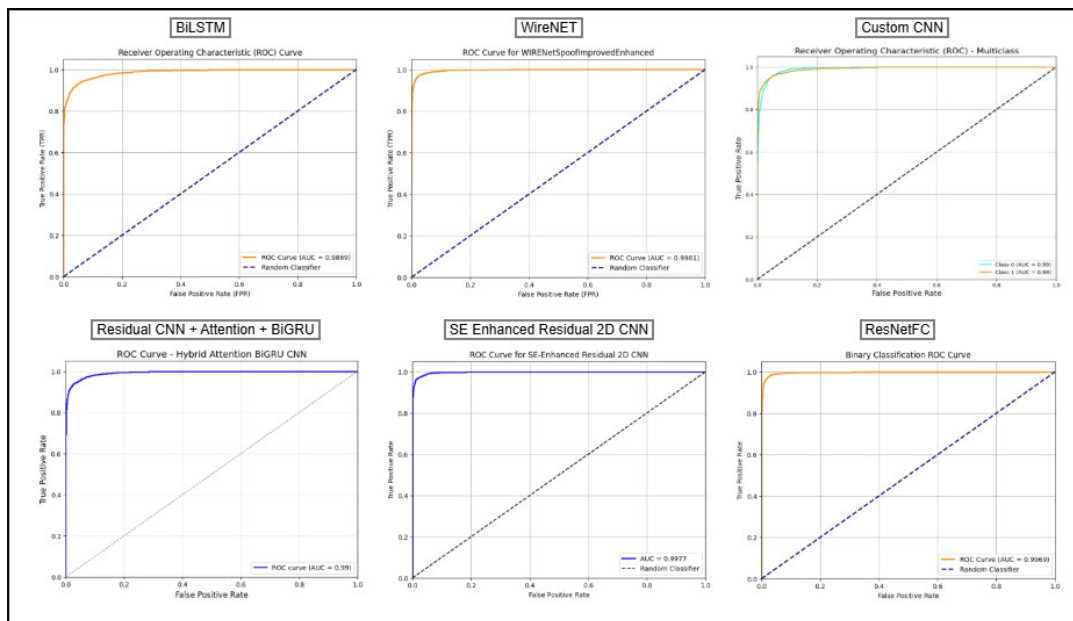


FIGURE 13. ROC curves of the proposed models for the feature Set-02.

A fully connected neural network, on the other hand, yielded a comparatively lower accuracy of 85%. Notably, the proposed SE-Enhanced 1D-CNN outperformed existing methods, achieving the higher accuracy of 98%, thereby highlighting its robustness in audio deepfake detection.

G. LIMITATIONS AND FUTURE WORKS

The proposed deep learning approaches demonstrate promising results on ASVspoof 2019 dataset, but there

remain several limitations. First, the generalization capability of current models to unseen spoofing techniques is still inadequate, and deepfake detection remains highly dataset-dependent and vulnerable to domain shifts. Second, the robustness of detection methods in real-world scenarios is not fully established. Factors such as background noise, video compression, or adversarial perturbations may compromise reliability in practice. Moreover, most state-of-the-art models require high computational cost, which can hinder deployment in real-time systems. Finally, ethical and

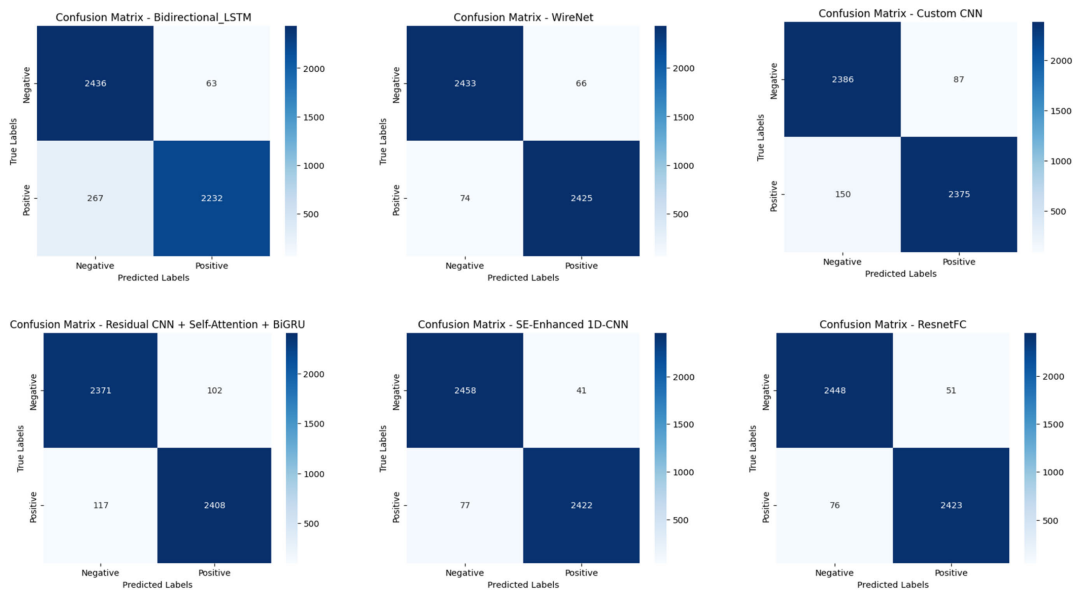


FIGURE 14. Confusion matrices of the proposed models for the feature Set-03.

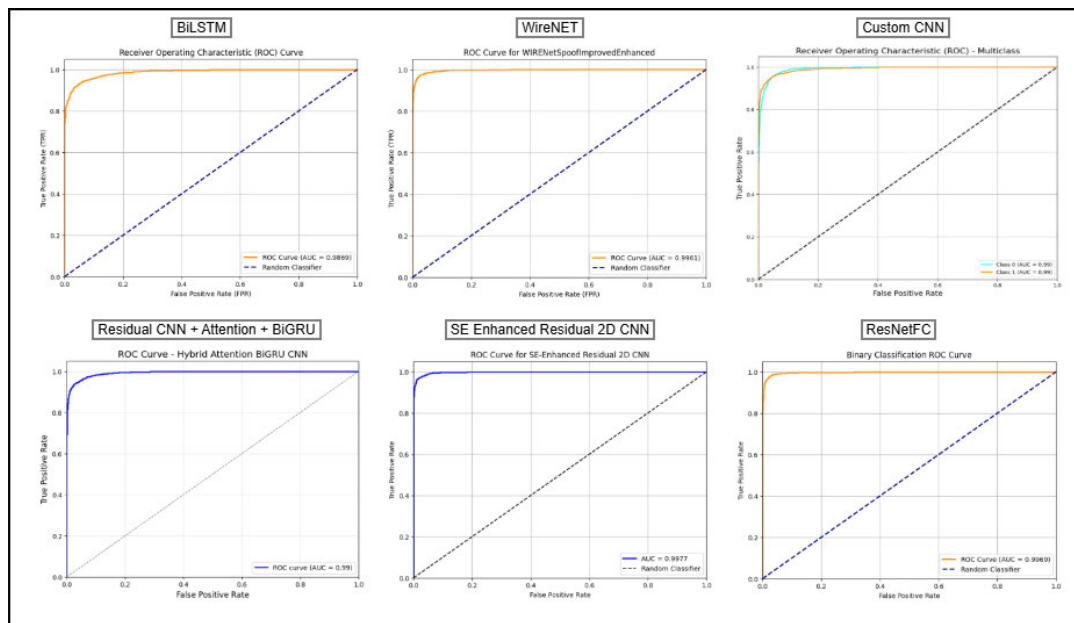


FIGURE 15. ROC curves of the proposed models for the feature Set-03.

societal challenges persist, with the continuous evolution of generative models creating an arms race between creation and detection, while issues related to privacy, fairness, and responsible use of detection technologies remain open for future research. One limitation of this study is that adversarial robustness was not explicitly evaluated.

In future work, the robustness of our approach under various adversarial scenarios, including perturbation-based attacks and more sophisticated manipulation strategies, to ensure its reliability in real-world applications. Addi-

tionally, other approaches can be applied to multilingual datasets, such as ASVspoof 2021, to quantify the dependency of language and enhance cross-linguistic robustness, and to apply pruning techniques to improve computational efficiency and enable exploration of lightweight deployment strategies without compromising detection performance. Also, future advancements in DL techniques hold the potential to enhance deepfake audio detection, making such systems more resilient, interpretable in safeguarding against manipulated audio.

TABLE 5. Performance of models utilizing only MFCC vs. MFCC + extra features.

Model	Feature Set	Accuracy	Precision	Recall	F1-Score
BiLSTM	Only MFCC	0.870	0.870	0.870	0.870
	Extra Features	0.940	0.941	0.940	0.940
WIRENET	Only MFCC	0.960	0.959	0.960	0.960
	Extra Features	0.970	0.973	0.970	0.970
ResNet FC	Only MFCC	0.962	0.962	0.962	0.962
	Extra Features	0.976	0.976	0.976	0.976
SE-Enhanced 2D CNN	Only MFCC	0.947	0.946	0.947	0.945
	Extra Features	0.975	0.975	0.975	0.975
Custom CNN	Only MFCC	0.917	0.918	0.917	0.915
	Extra Features	0.948	0.949	0.948	0.950
ResCNN + Attention BiGRU	Only MFCC	0.950	0.950	0.950	0.950
	Extra Features	0.960	0.960	0.960	0.960

TABLE 6. Ablation study results for SE enhanced 1D CNN.

Model Configuration	Accuracy	F1 Score	Precision	Recall	Impact vs Full Model
Baseline (Full Model)	0.9650	0.9646	0.9654	0.9650	Reference
Shallow Model	0.9386	0.9369	0.9404	0.9386	-0.0264 (-2.74%)
No Batch Normalization	0.9495	0.9485	0.9504	0.9495	-0.0155 (-1.61%)
No SE Attention	0.9537	0.9533	0.9540	0.9537	-0.0113 (-1.17%)
No Residual Connections	0.9546	0.9542	0.9546	0.9546	-0.0104 (-1.08%)
No L2 Regularization	0.9615	0.9609	0.9624	0.9615	-0.0035 (-0.36%)
Standard Loss (No Focal)	0.9640	0.9635	0.9644	0.9640	-0.0010 (-0.10%)
No Dropout	0.9788	0.9787	0.9788	0.9788	+0.0138 (+1.43%)

TABLE 7. Comparison with the existing literature on the ASVspoof 2019 dataset.

Model	Accuracy	References
SVM	92.89%	[33]
CNN+BiLSTM	97.21%	[34]
SVM	89%	[35]
Xception model	92.11%	[36]
Fully connected neural network	85%	[37]
SE-Enhanced 1D-CNN	98%	Proposed model

H. ETHICAL IMPLICATIONS OF DEEFAKE DETECTION

The ethical concerns for the deepfake detection technology are essential. The sensitivity surrounding the use of deepfake detection tools, particularly issues of misuse, false positives in legal contexts, and surveillance. It involves several critical risks like governmental or selective censorship, potential undermining of authentic audio, and the catastrophic consequences of both false positives and false negatives. Notably, the mention of false positives discrediting valid evidence in journalistic contexts and false negatives enabling misinformation highlights a balanced understanding of the technology's dual-edge risks. Also, surveillance risks—specifically the potential repurposing of detection techniques for invasive monitoring—are well-founded. The ethical consideration can be taken as concrete frameworks for ethical deepfake detection deployment, such as guidelines on explainability, data governance, and stakeholder engagement. Addressing algorithmic fairness and inclusivity is critical for social acceptance and legitimacy. Elaborating on the interplay

between technology and existing legal frameworks governing digital evidence, privacy, and freedom of expression would add depth. Also, need to highlight the role of public education about the limitations of detection technologies can mitigate undue reliance on imperfect systems, reducing harms from misinterpretation.

VI. CONCLUSION

The manipulation of audio using AI techniques, particularly DL, has become a critical societal concern. Malicious actors exploit deepfake technologies to generate deceptive content, posing significant threats to political stability, security, and personal privacy. This study evaluates the effectiveness and robustness of the DL models in detecting deepfake audio, demonstrating that the SE-Enhanced 1D-CNN model achieves the highest accuracy of 98%, surpassing BLSTM, WIREnet, SVM, ResNet FC, custom CNN, and a residual CNN with attention-enhanced BGRU. We also observed that the performance of the models varies depending on the feature representations generated from the input signal. Therefore, selecting the appropriate temporal and spectral features from real and fake audio is crucial for distinguishing between them. The results underscore the urgency of developing robust detection mechanisms to counteract the increasing sophistication of deepfake-generated speech. The implementation of the proposed work is provided in the following GitHub repository: Code: Audio-Deep-Fake-Comparison.

CONFLICTS OF INTEREST

No conflicts of interest are declared by the authors.

DATA AVAILABILITY

For this study, the dataset utilized is sourced from publicly available open data repositories. Additional details can be obtained upon reasonable request from the corresponding author.

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